ChE 210 Homework Set #9 Due on Friday, 14 November 2025

(1) Murphy 5.40

(2) Murphy 5.41 The figure included in the problem statement shows c_p in mmol/L plotted on the horizontal axis and q_p in mmol/g plotted on the vertical axis. The data are given at the right. Fit these data to two adsorption models: (1) the Langmuir isotherm:

$$q_p = q_\infty \frac{K_L\, c_p}{1+K_L\, c_p}$$

where q_∞ and K_L are adjustable parameters and (2) the Freundlich isotherm:

$$q_p = k_F\, c_p^{1/n}$$

where k_F and n are another set of adjustable parameters. Try fitting the data to both models. Which is better? Use that model to solve the problem.

cp (mmol/L)	qp (mmol/g)
0.02	0.79
0.11	1.25
0.36	1.65
0.43	1.86
0.60	1.88
0.89	2.11
0.98	2.18
1.81	2.56
2.30	2.72
3.06	2.79
3.59	2.90
4.05	3.01
4.71	2.99
5.27	3.17
6.05	3.21
6.69	3.30

(3) Air is heated from 25°C to 150°C prior to entering a combustion furnace. The change in specific enthalpy associated with this transition is 3640 J/mol. The flow rate of air at the heater outlet is 1.25 m^3/min and the air pressure at this point is 122 kPa absolute.

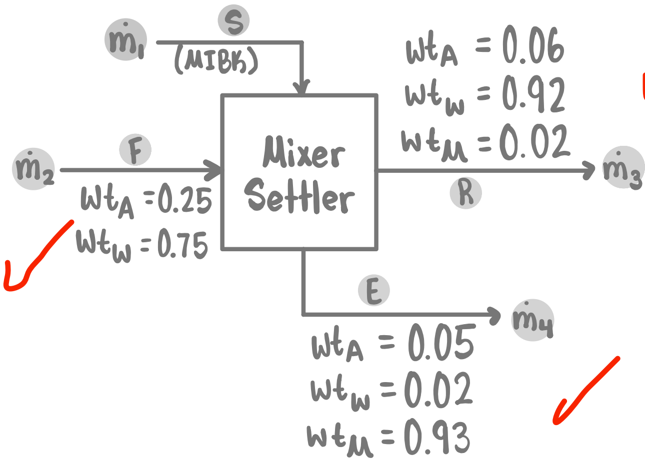
(a) Calculate the heat requirement in kW, assuming ideal gas behavior and that the kinetic and potential energy changes from the heater inlet to the outlet are negligible.

(b) Would the value of $\Delta\dot{E}_k$ [which was neglected in part (a)] be positive or negative, or would you need more information to answer? If you need more information, what would that information be?

(4) Murphy 6.26

1.

P5.40 Acetic acid is to be extracted from water in a single mixer-settler unit using pure methyl isobutyl ketone (MIBK) as the solvent. The feed contains 25 wt% acetic acid in water, and the water-rich product stream must contain no more than 6 wt% acetic acid. Calculate the kg solvent/kg feed and the compositions and flow rates of the two streams leaving the process. (Refer to Fig. 5.15.)



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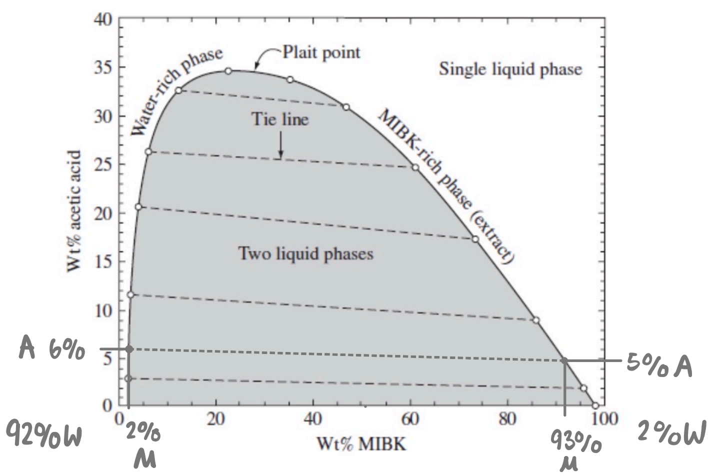


Figure 7.13 MIBK-acetic acid-water liquid-liquid phase diagram at 25°C. Drawn from data in *Perry's Chemical Engineers' Handbook,* 6th edition.

Composition Summary

	A	M	W
$\dot m_3$ R	0.06	0.02	0.92
$\dot m_4$ E	0.05	0.93	0.02

Species Balance

(In) = (Out)

A $0.25\dot m_1 = 0.06\dot m_3 + 0.05\dot m_4$

W $0.75\dot m_1 = 0.92\dot m_3 + 0.03\dot m_4$

M $1.00\dot m_2 = 0.02\dot m_3 + 0.93\dot m_4$

if set basis of $\dot m_1 = 100$ kg/hr

$$\begin{pmatrix} 0.06 & 0.05 \\ 0.92 & 0.03 \end{pmatrix}\begin{pmatrix} \dot m_3 \\ \dot m_4 \end{pmatrix} = \begin{pmatrix} 25 \\ 75 \end{pmatrix}$$

$\dot m_3 = 67.873$ kg/hr

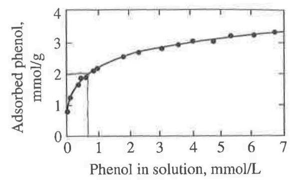
$\dot m_4 = 418.552$ kg/hr

$\dot m_2 = (67.873)(0.02) + (418.542)(0.93)$

$\dot m_2 = 390.611$ kg/hr

$$\frac{\text{kg Solvent}}{\text{kg Feed}} = \frac{(\dot m_2)}{(\dot m_1)} = \frac{390.611}{100} = \boxed{3.91\ \frac{\text{kg Solvent}}{\text{kg Feed}}}$$

P5.41 Trace organic contaminants in waste water readily adsorb onto a solid called activated carbon which is not very different from charcoal briquets. Equilibrium data for adsorption of phenol onto activated carbon is shown. One liter of wastewater containing 1 g phenol is contacted with 5 g activated carbon at 20°C. At equilibrium, what fraction of the phenol is adsorbed?



P NOT Abs | P Abs

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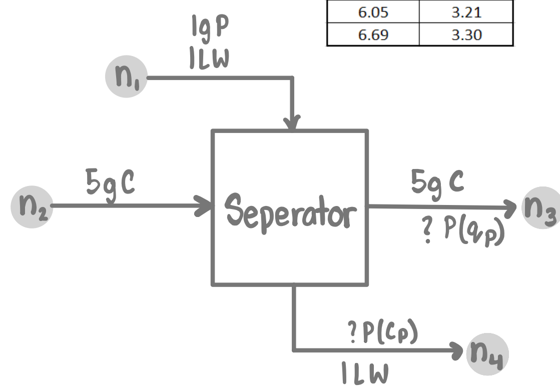
2.

$$\frac{1\,g\,Phenol}{1L} \times \frac{10^3\,mg}{1g} \times \frac{1\,mmol\,P}{94.11\,mg\,P} = 10.63\,mmol\,Phenol$$



Langmuir Isotherm

$$q_p = q_\infty\left(\frac{K_L C_p}{1+K_L C_p}\right)$$

$$y = a\left(\frac{bx}{1+bx}\right)$$

$$\frac{1}{y} = \frac{1+bx}{abx}$$

$$\frac{1}{y} = \frac{1}{ab}\left(\frac{1}{x}\right) + \frac{1}{a}$$

$$y = m(x) + b$$

Freundlich Isotherm

$$q_p = K_F C_p^{1/n}$$

$$y = cx^{1/d}$$

$$\log(y) = \log(cx^{1/d})$$

$$\log(y) = \log(x^{1/d}) + \log(c)$$

$$\log(y) = \frac{1}{d}\log(x) + \log(c)$$

$$y = m(x) + b$$

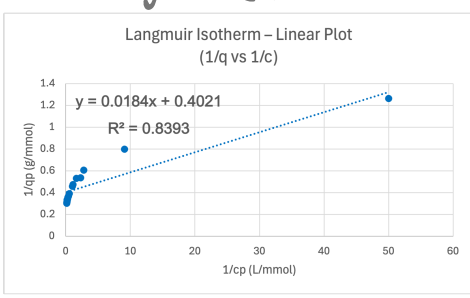
Mole Balance

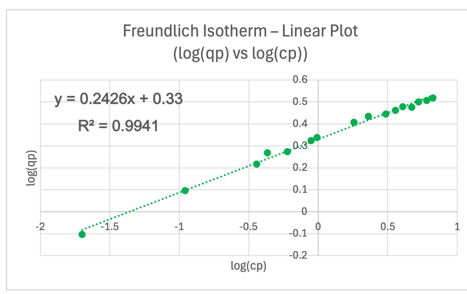
$$n_{1,p} = n_{4,p} + n_{3,p}$$

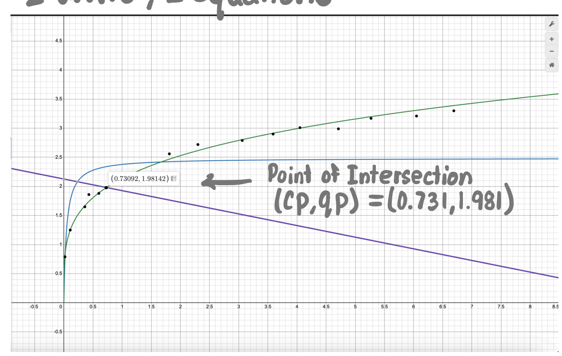
$$10.63\,mol\,P = cp(1L) + qp(5g)$$

$$q_p = (2.138)\,cp^{1/4.122}$$

2 Unks, 2 Equations



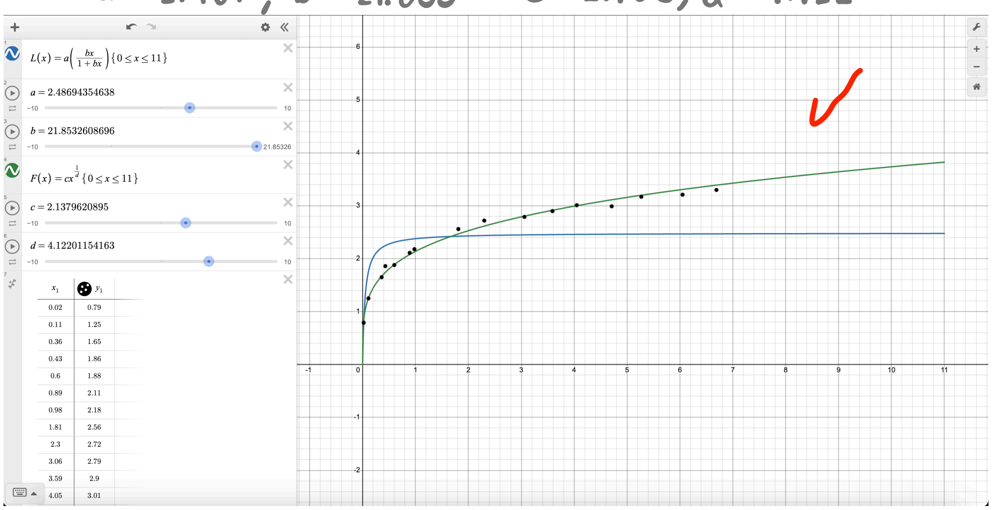




Point of Intersection $(cp, qp) = (0.731, 1.981)$

$$\frac{1}{ab} = 0.0184, \quad \frac{1}{a} = 0.4021 \qquad \frac{1}{d} = 0.2426 \quad \log(c) = 0.33$$

$$a = 2.487, \; b = 21.853 \qquad c = 2.138, \; d = 4.122$$



$$f^P_{Abs} = \frac{n_{P,Abs}}{n_{P,in}} = \frac{n_{3,P}}{n_{1,P}}$$

$$f^P_{Abs} = \frac{(1.981\,mmol/g\,Phenol)(5g)}{10.63\,mmol\,Phenol}$$

$$f^P_{Abs} = 0.9320 \text{ or } \boxed{93.20\%}$$

Freundlich Isotherm model has a better fit using linear regression because the linearity coefficient is closer to 1. Also graphically the graph has a better fit to the data.

$$q_p = (2.487)\left(\frac{21.853}{1+21.853C_p}\right) \qquad \boxed{q_p = (2.138)C_p^{1/4.122}}$$

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3.

(3) Air is heated from 25°C to 150°C prior to entering a combustion furnace. The change in specific enthalpy associated with this transition is 3640 J/mol. The flow rate of air at the heater outlet is 1.25 m³/min and the air pressure at this point is 122 kPa absolute.

(a) Calculate the heat requirement in kW, assuming ideal gas behavior and that the kinetic and potential energy changes from the heater inlet to the outlet are negligible.

(b) Would the value of $\Delta \dot{E}_k$ [which was neglected in part (a)] be positive or negative, or would you need more information to answer? If you need more information, what would that information be?

$\Delta H = 3640$ J/mol

$P = 122$ KPa

$\dot{V} = \frac{1.25\,m^3}{min}$

A. $n = \frac{PV}{RT} = \frac{(122 \times 10^3\,\cancel{Pa})\left(1.25\frac{\cancel{m^3}}{\cancel{min}} \times \frac{1\,\cancel{min}}{60\,s}\right)}{\left(8.314\frac{\cancel{Pa}\cdot\cancel{m^3}}{mol\cdot\cancel{K}}\right)(423.15\,\cancel{K})} = 0.722\,\frac{mol}{s}$

$\frac{kJ}{s}$ [=] kw

1st Law: $\Delta\dot{H} + \cancel{\Delta\dot{E}_k}^{\,0} + \cancel{\Delta\dot{E}_p}^{\,0} = \dot{Q} + \cancel{\dot{W}_s}^{\,0}$

$\therefore \dot{Q} = \Delta\dot{H} = \dot{n}\Delta\hat{H} = \left(0.722\,\frac{mol}{s}\right)\left(3640 \times 10^{-3}\,\frac{kJ}{mol}\right) = 2.63\,\frac{kJ}{s} = \boxed{2.63\text{ kw}}$

B. I would need more information in order to determine sign of ΔE_k. As the calculation for ΔE_k involves mass and velocity. Which no velocity was given.

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P6.26 Calculate the energy change, in J, for 1 gmol of H_2O undergoing the following changes:

(a) accelerating from 0 to 100 km/h
(b) rising 100 meters
(c) increasing in pressure at 100°C from 0.1 bar to 1 bar
(d) increasing in temperature from 0 to 100°C at 5 bar
(e) dissociating to H_2 and O_2 at 25°C
(f) undergoing a nuclear reaction where all mass converts to energy ($E = mc^2$)

$1\,\text{gmol } H_2O = \frac{18.01\,g\,H_2O}{1\,\text{gmol } H_2O}$

4.

A. $\Delta E_k = \frac{1}{2} m (v_f^2 - v_i^2) = \frac{1}{2}\left(1\,\text{gmol } H_2O \times \frac{18.01\,g\,H_2O}{1\,\text{gmol } H_2O}\right)\left(\left(100\,\frac{km}{h}\right)^2 - \left(0\,\frac{km}{h}\right)^2\right)\left(\frac{1\,m}{10^{-3}\,km}\right)^2\left(\frac{1\,h}{3600\,s}\right)^2\left(\frac{1\,kg}{10^3\,g}\right)$

$\Delta E_k = 6.948\,\frac{kg \cdot m^2}{s^2} = \boxed{6.948\,J}$ ✓

B. $\Delta E_p = mg(z_f - z_i) = (18.01\,g\,H_2O)\left(9.81\,\frac{m}{s^2}\right)(100\,m - 0\,m)\left(\frac{1\,kg}{1000\,g}\right)$

$\Delta E_p = 17.668\,\frac{kg \cdot m^2}{s^2} = \boxed{17.668\,J}$ ✓

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C. $\Delta U = m\,\Delta\hat{U} = (18.01\,g\,H_2O)(2506.2 - 2515.5)\,\frac{J}{g} = \boxed{-167.493\,J}$ ✓

$\hat{U}$ from (pg 513) $\hat{U}(100°C, 0.1\,\text{barr}) = 2506.2\,J/g$
$\hat{U}(100°C, 1\,\text{barr}) = 2515.5\,J/g$

D. $\Delta U = n C_v \Delta T = (1\,\text{gmol } H_2O)\left(75.4\,\frac{J}{mol\,°C}\right)(100-0)\,C° = \boxed{7540\,J}$ ✓

($C_v \approx C_p$ from pg. 666 Murpy)

E. $H_2O \longrightarrow H_2 + \frac{1}{2}O_2$

$\Delta\hat{H}_r = (0)(0) + (0)(0) - (1)\left(-285.85\,\frac{kJ}{gmol}\right) = 285.85\,\frac{kJ}{gmol}$

$\Delta\hat{U}_r = \Delta\hat{H}_r - RT\Delta n_r = \left(285.85\,\frac{kJ}{gmol}\right)\left(\frac{1\,J}{10^{-3}\,kJ}\right) - \left(8.314\,\frac{J}{mol \cdot K}\right)(298\,K)\left(\frac{1}{2}\,gmol\right) =$

$\boxed{\Delta\hat{U}_r = 284{,}611\,J}$ ✓

F. $\Delta E = (18.01\,g\,H_2O)\left(2.99792 \times 10^8\,\frac{m}{s}\right)^2\left(\frac{1\,kg}{10^3\,g}\right) = \boxed{1.619 \times 10^{15}\,J}$ ✓